

Customer Review Analysis - Amazon

Group 7

Division of work

- NLP and environment setup - M11351002陳濤恩
- NAPL Product preferences, sentiment analysis, keywords analysis - B11001019傅煒宸, B11001137許庭瑄
- Temporal review trend for popular products, bundle product - M11351015李國禎
- Random forest, keywords in reviews, - M11301302黃冠翔
- Slide, presentation - M11301108周易儒

Analysis Framework & Overview

- MongoDB Data Collection
- NLP Preprocessing (Tokens, Cleaning, Sentiment...)
- Customer Preference Analysis
- Sentiment Analysis
- Time Series & Model Validation (Random Forest)
- Keywords in the reviews
- Business Recommendations & Platform Optimization

Data Collection & NLP Pipeline

- Download review dataset, ingest into MongoDB, and create indexes.
- Perform NLP tasks and enrich documents with structured linguistic features. Ex: Tokens, lemmas, Sentiment score...
- Begin Analysis

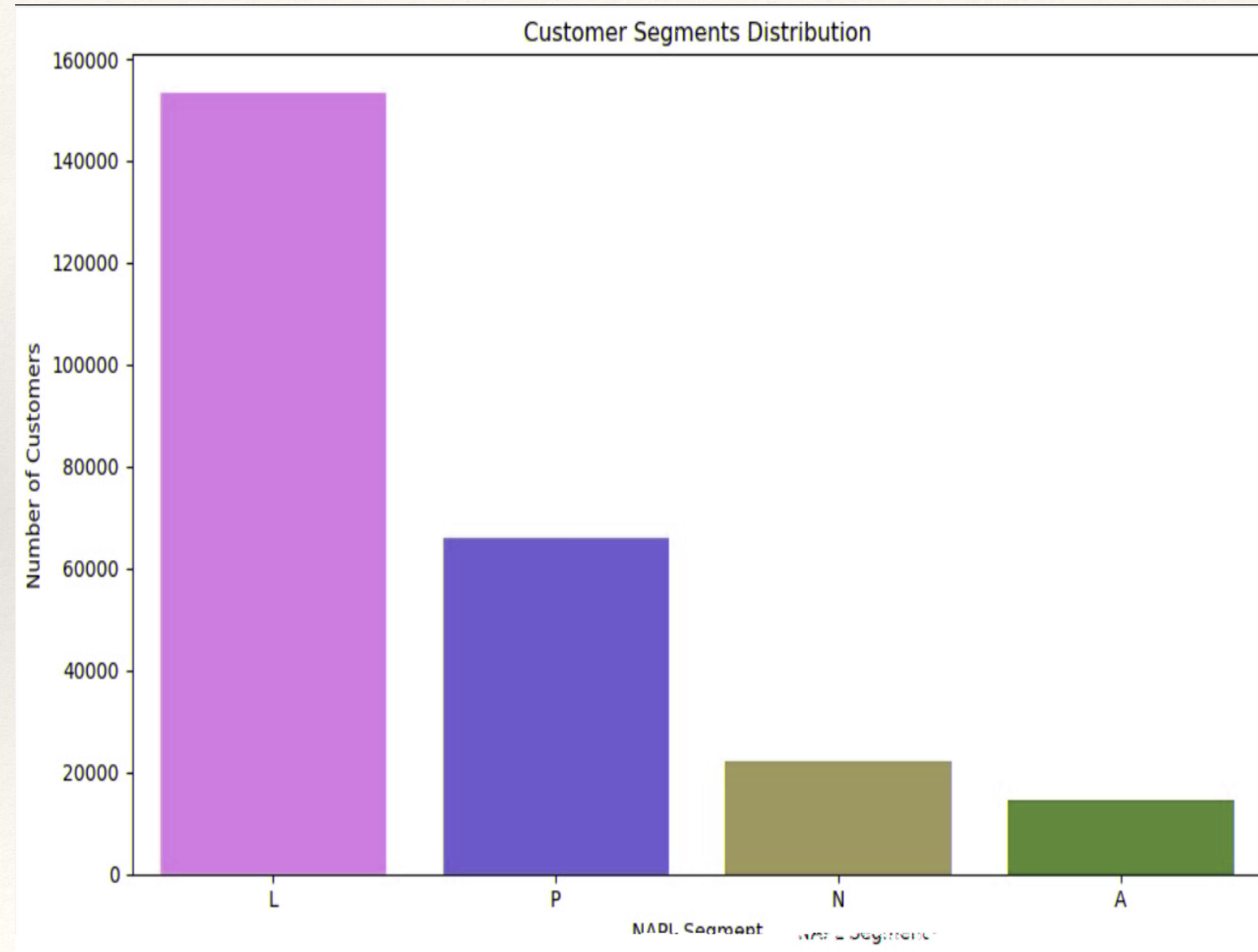
```
{
  _id: ObjectId,           // MongoDB 自動產生的唯一識別碼
  ProductId: String,       // 產品 ID
  UserId: String,          // 使用者 ID
  ProfileName: String,     // 使用者名稱
  HelpfulnessNumerator: Number (Integer), // 認為評論有幫助的人數
  HelpfulnessDenominator: Number (Integer), // 評論總共被評估為有幫助的人數
  Score: Number (Integer), // 評論分數 (例如 1-5)
  Time: Number (Integer/Timestamp), // 評論的時間 (Unix 時間戳記)
  Summary: String,         // 評論標題
  Text: String,            // 完整評論文字
  Keywords: Array (String), // 關鍵字列表 (字串陣列)
  Lemmas: Array (String),  // 詞元列表 (字串陣列)
  Named_entities: Array (Array (String)), // 命名實體列表 (二維字串陣列, 例如 [['Vitality', 'ORG']])
  POS_tags: Array (Array (String)), // 詞性標籤列表 (二維字串陣列, 例如 [['buy', 'VERB']])
  Sentiment_score: Number (Float), // 情感分數 (浮點數)
  Text_cleaned: String,    // 清理後的評論文字
  Tokens: Array (String),  // 分詞列表 (字串陣列)
  Tokens_processed: Array (String) // 處理後的分詞列表 (字串陣列)
}
```

Collection: reviews

NAPL

NAPL - A model based on transaction data.

- N(new) - Purchased in last 3 months, first-time
- A(active) - Purchased in last 3 months, not first-time
- P(potential) - Purchased before, inactive for 3 months
- L(lost) - Only one purchase, inactive over 3 months

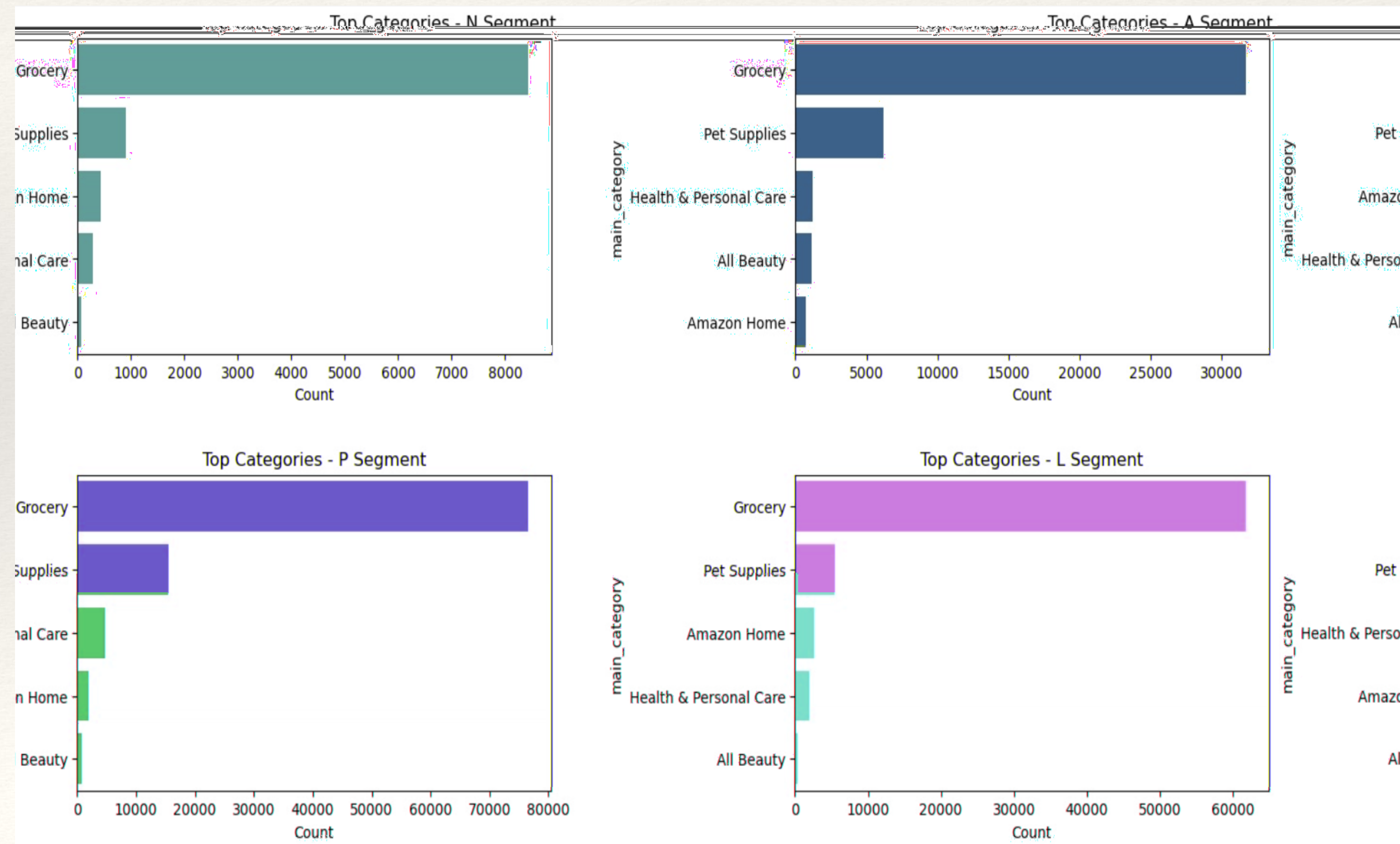


NAPL - Product Preferences

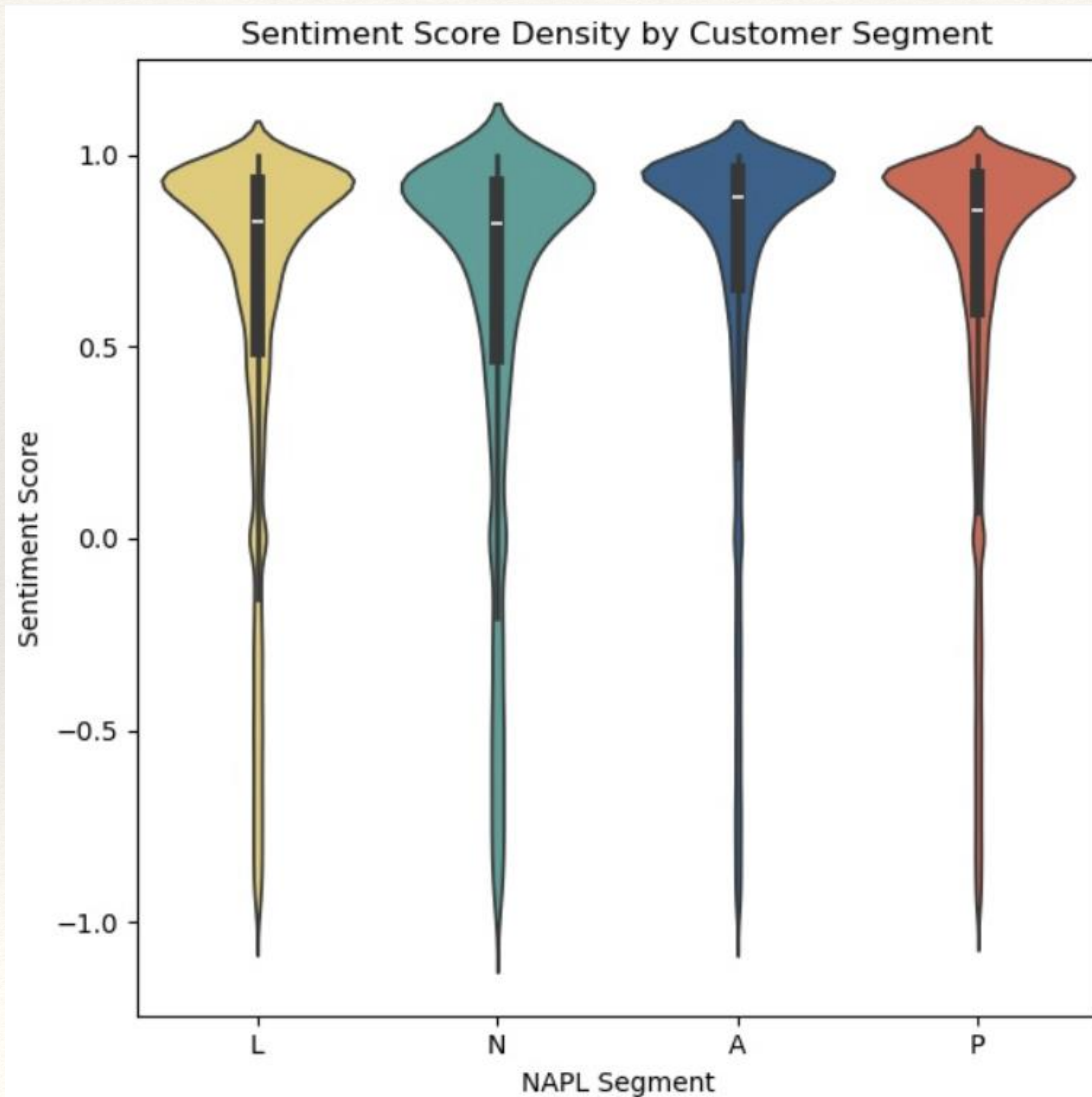
The top five categories of different groups are exactly the same, only the rankings are different. Grocery clearly takes the first place in every segment.

Suggest:

The platform can strengthen the recommendation system, promotional activities and service optimization in the Grocery and Pet Supplies categories.



NAPL - Sentiment Analysis

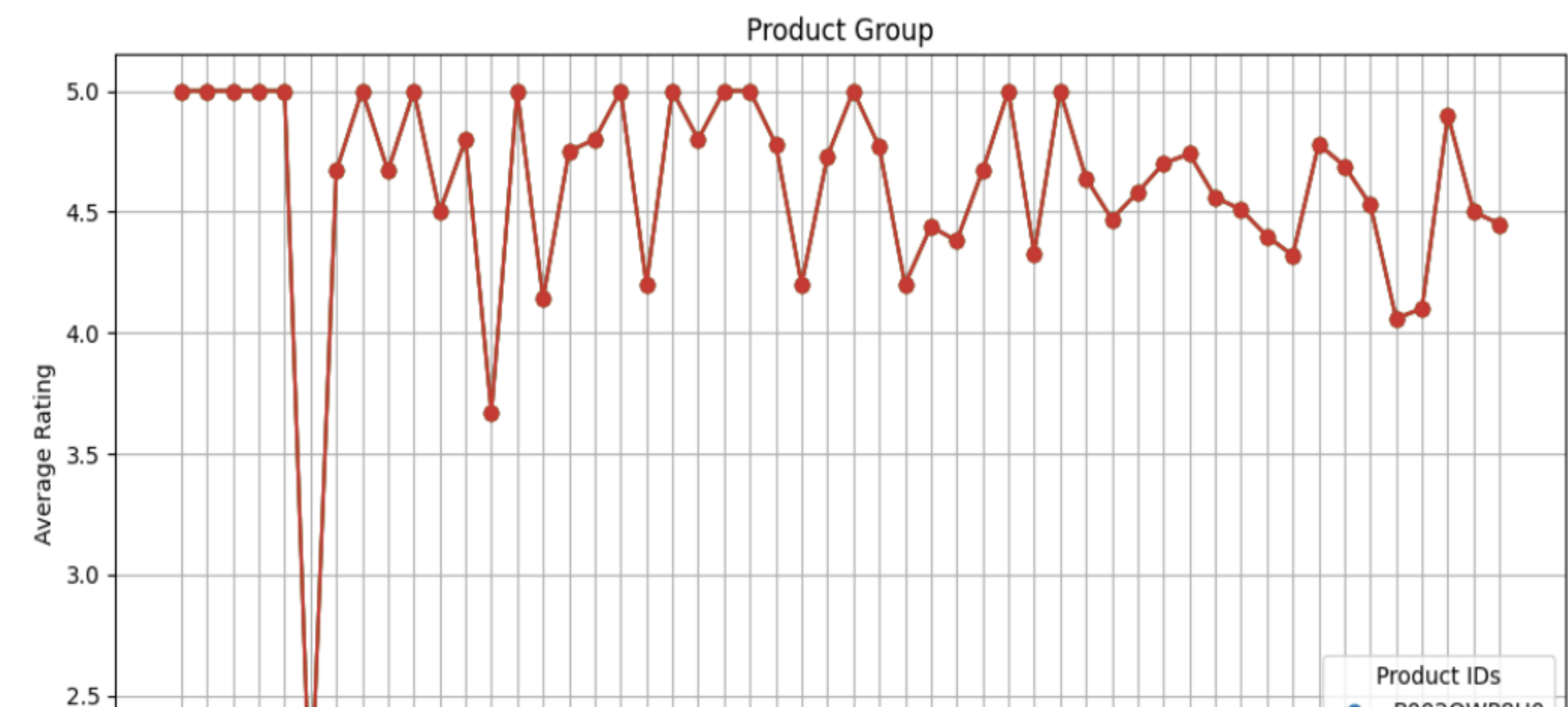


Violin plots

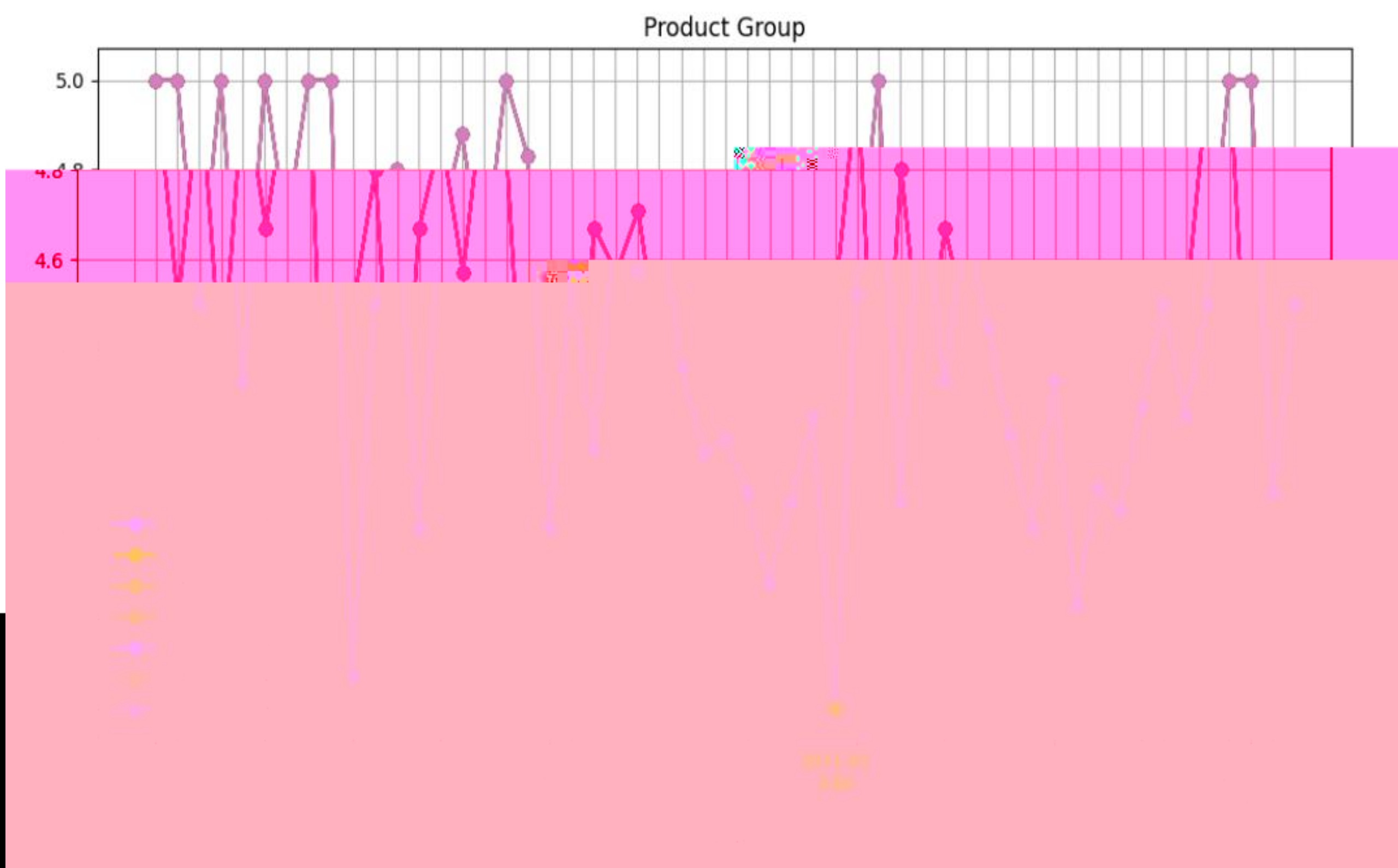
- Most customers fall in the 0.8 to 1.0 sentiment range.
- The “P” segment sentiment is the most stable and concentrated, performing strong brand satisfaction.
- The “N” segment generally gives positive feedback, but they also exhibit more negative sentiment outliers compared to other segments.

Temporal - Bundle Product Review

Some products have the same trend in their monthly average scores. This suggests that the items may be sold as a bundle, resulting in duplicate customer reviews.

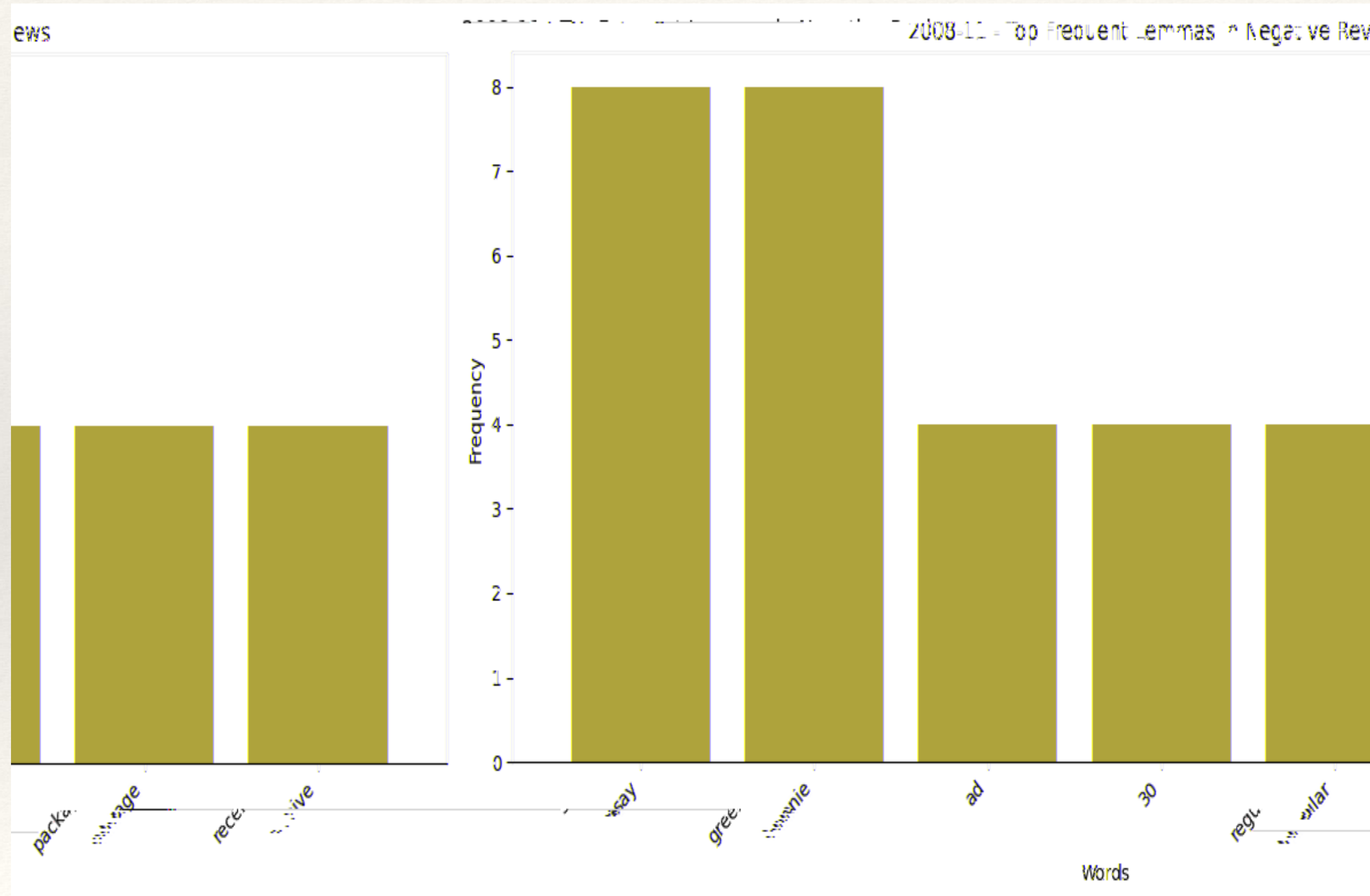


Plot1



Plot2

Temporal - Bundle Product Review

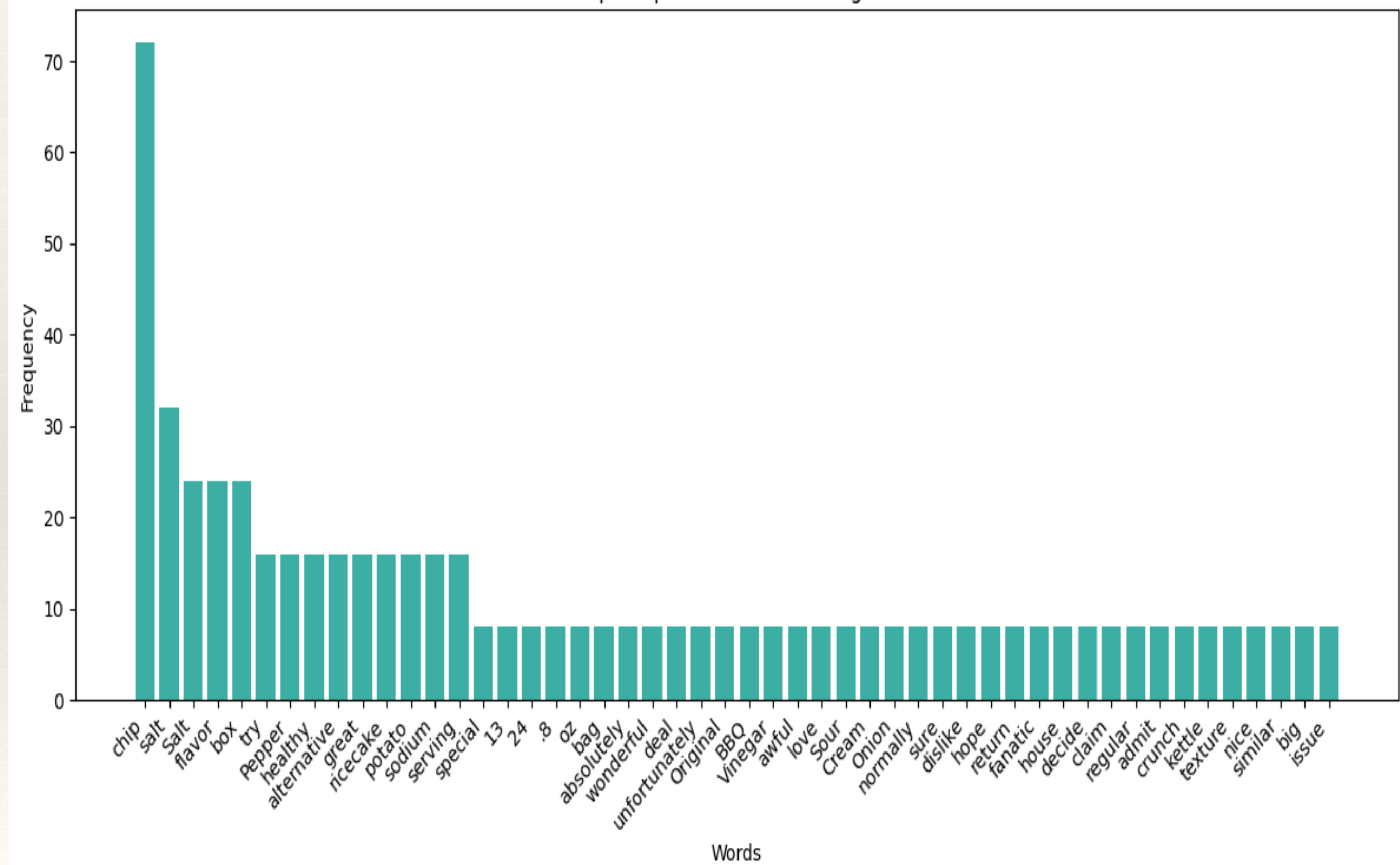


Greenies - Pet supplies

In the reviews for these 4 products, customers frequently mentioned that the advertised product contained 30 "greenies", but the packages received did not match the advertised description.

Temporal - Bundle Product Review

2011-01 - Top Frequent Lemmas in Negative Reviews



Popchips

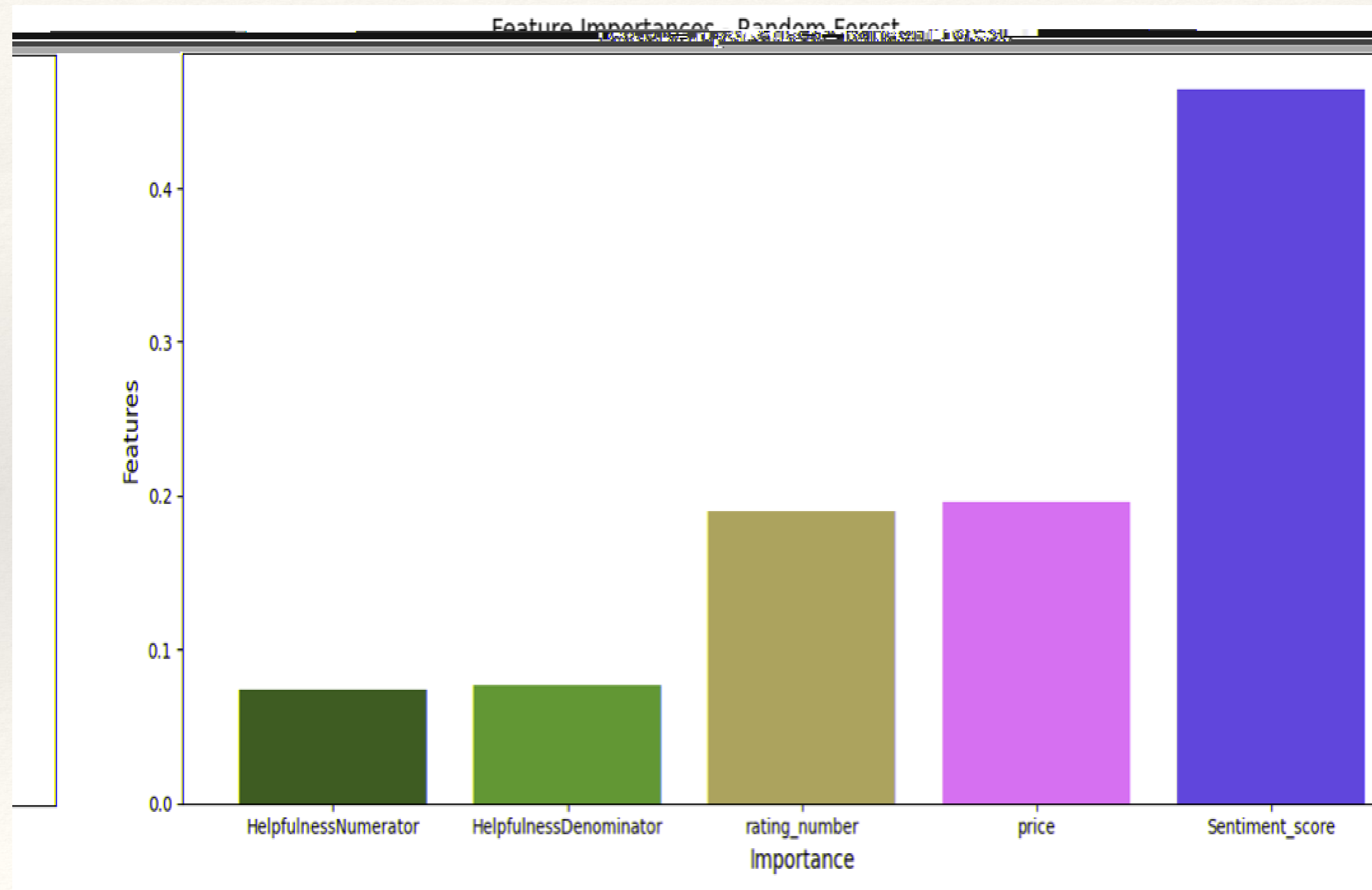
In the reviews for these 7 products, the reviews frequently mention overly salty flavor and damaged packaging.

Suggest: Optimize the combined product rating system to improve fairness and reference value.

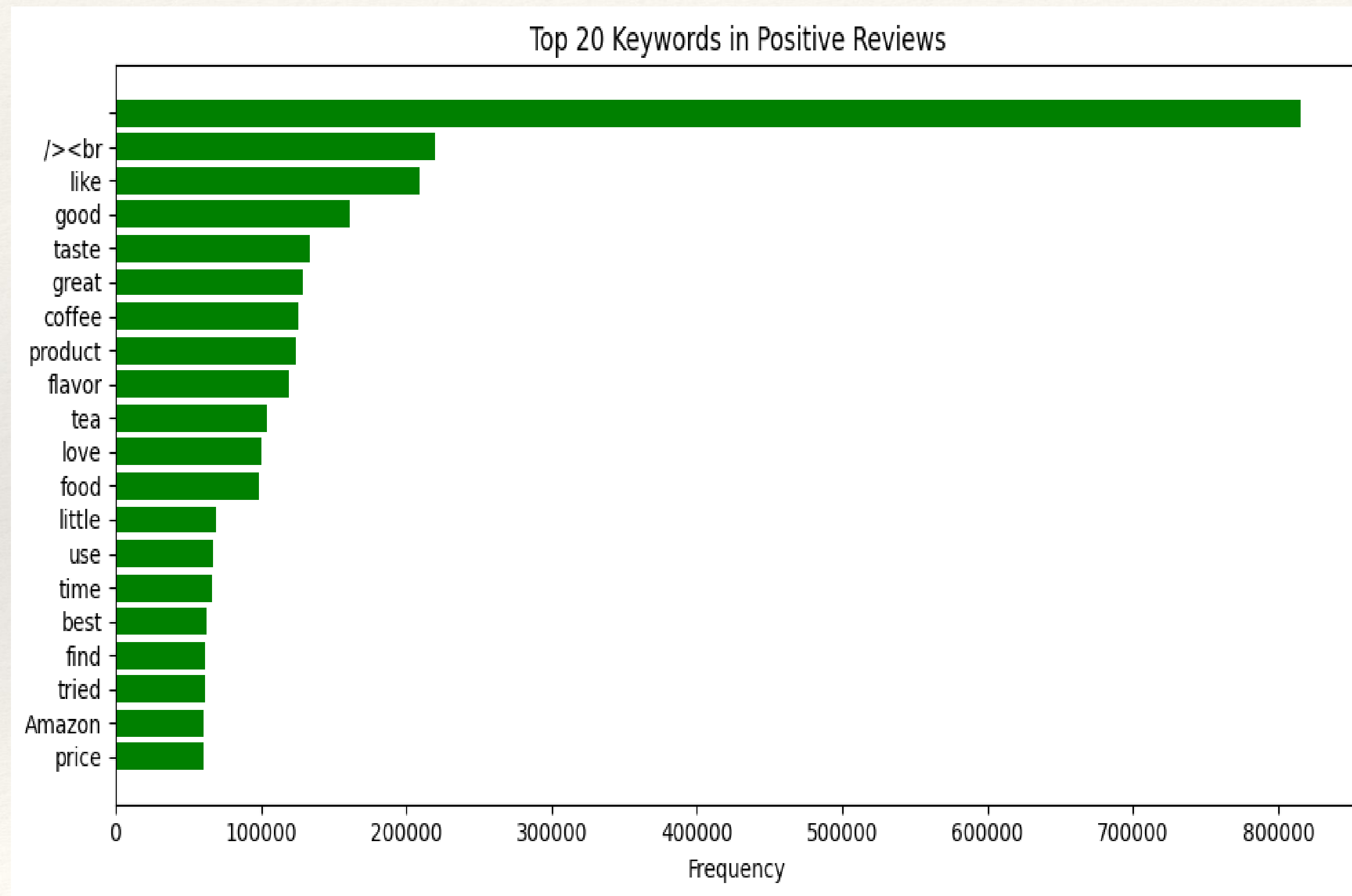
Does Sentiment Affect Ratings? - Random Forest Analysis

This shows that the emotions expressed by the consumer in text are highly consistent with the actual score they gave.

Suggest: Strengthen sentiment analysis for reviews and product recommendation.



Keywords in reviews - Positive



After excluding /><br, we can see many positive words related to food and beverages.

NAPL - Keywords in reviews - Positive

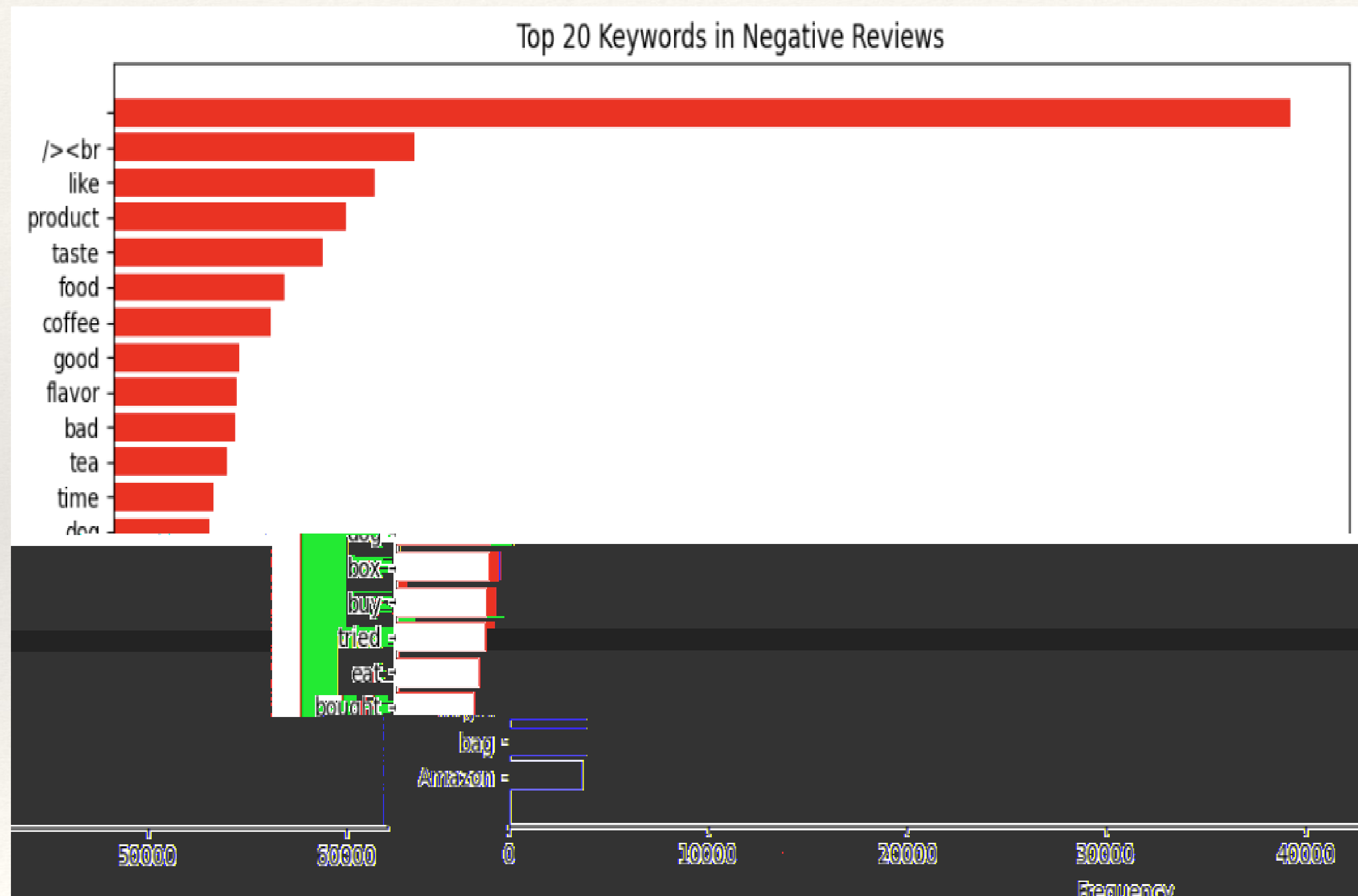


Focus on active customer, we observed that active customers attach more importance to "flavor" and "taste" than other.

Suggest: Optimizing product flavor will help improve satisfaction and loyalty for active customers .

Active customer

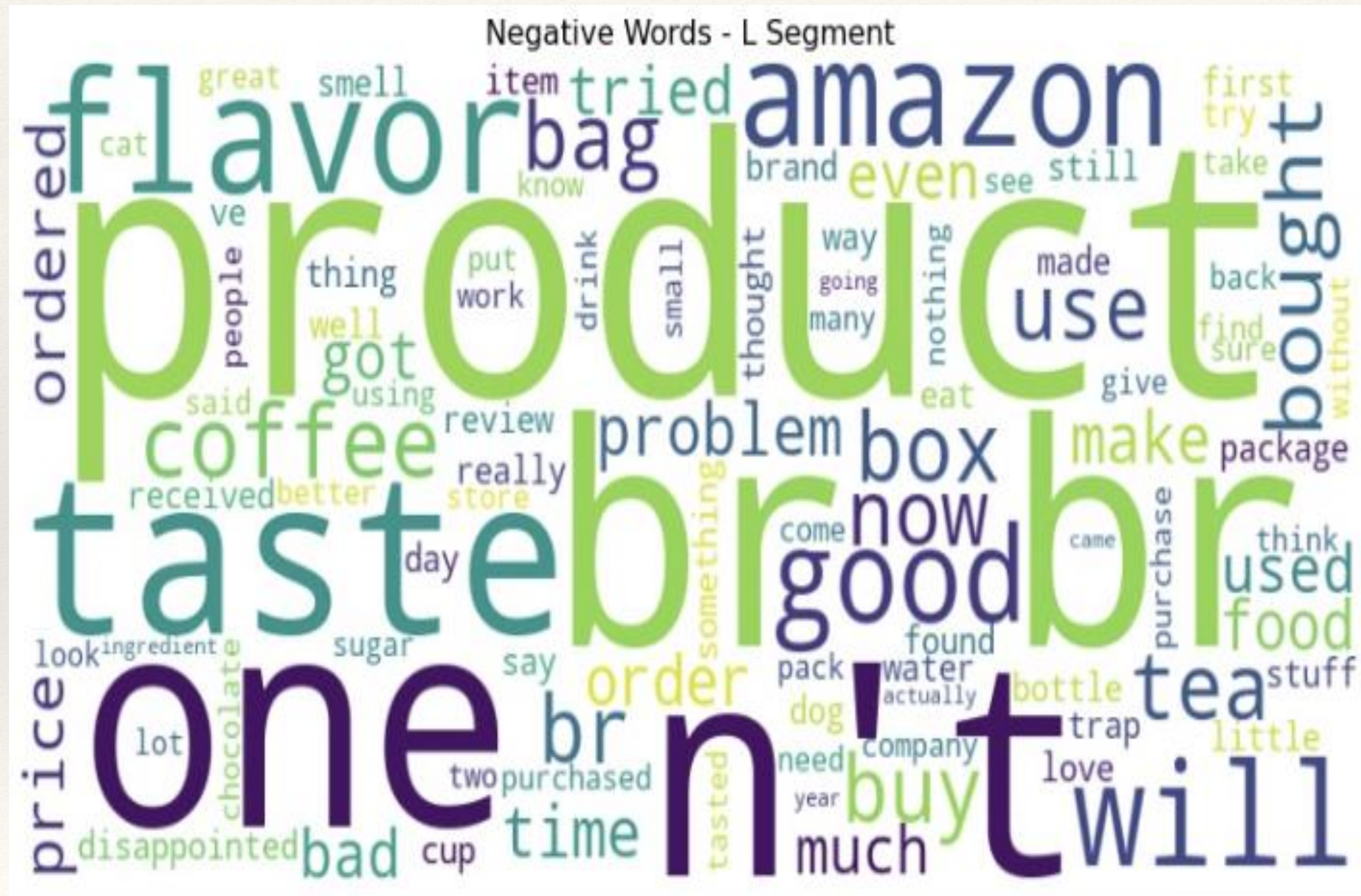
Keywords in reviews - Negative



Can see some common words, some of which also appear in positive reviews, but may have a negative tone in the context of negative reviews.

Suggest: When developing sentiment recognition models, pay attention to semantic context to avoid misinterpreting "not good" as a positive comment.

NAPL - Keywords in reviews - Negative



Lost customer

Focus on lost customer that is prone to churn.

Suggest: After analysis, there will have a basis for improvement.

Conclusion & Future

The previous analyses show that consumer rating behavior is influenced by multiple factors.

Future direction:

- Enhance recommendation systems by incorporating both sentiment analysis and customer preference data.
- Leverage analytical models to enable precision targeting and emotionally intelligent marketing.
- Develop next-level personalization for product recommendations.